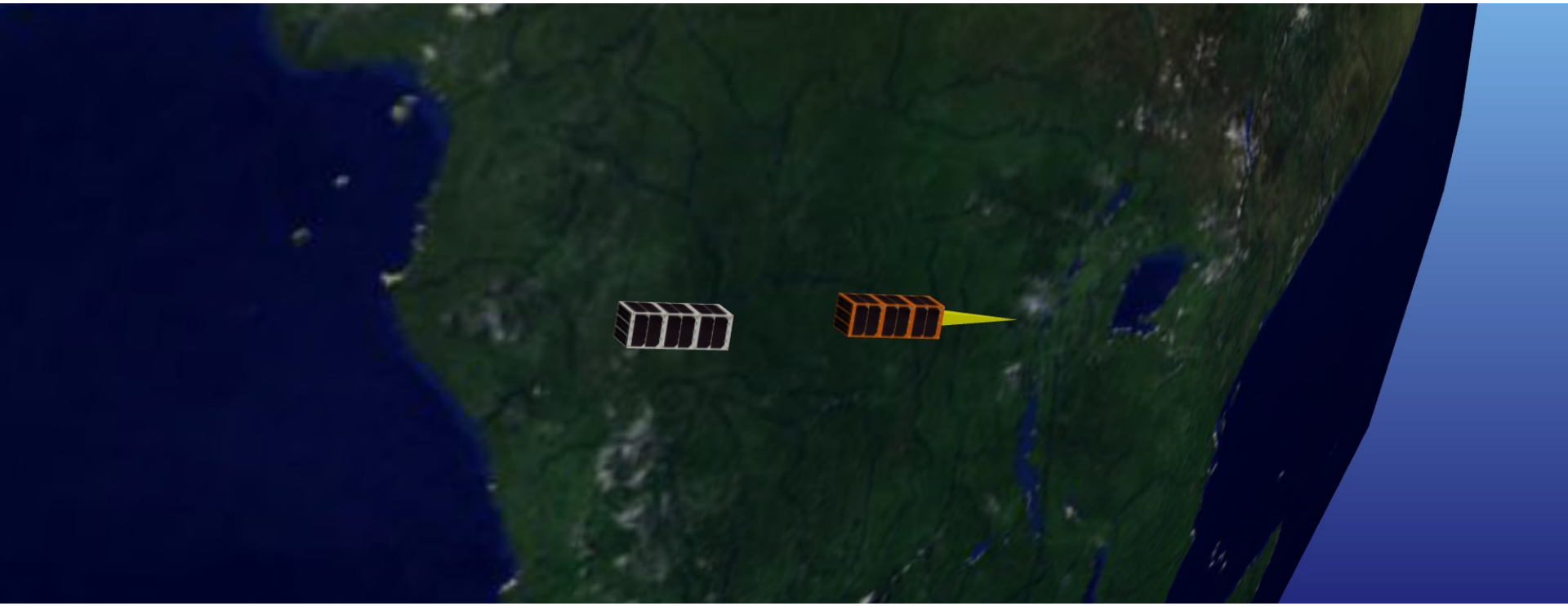
Fast Solution of Optimal Control Problems with L1 Cost



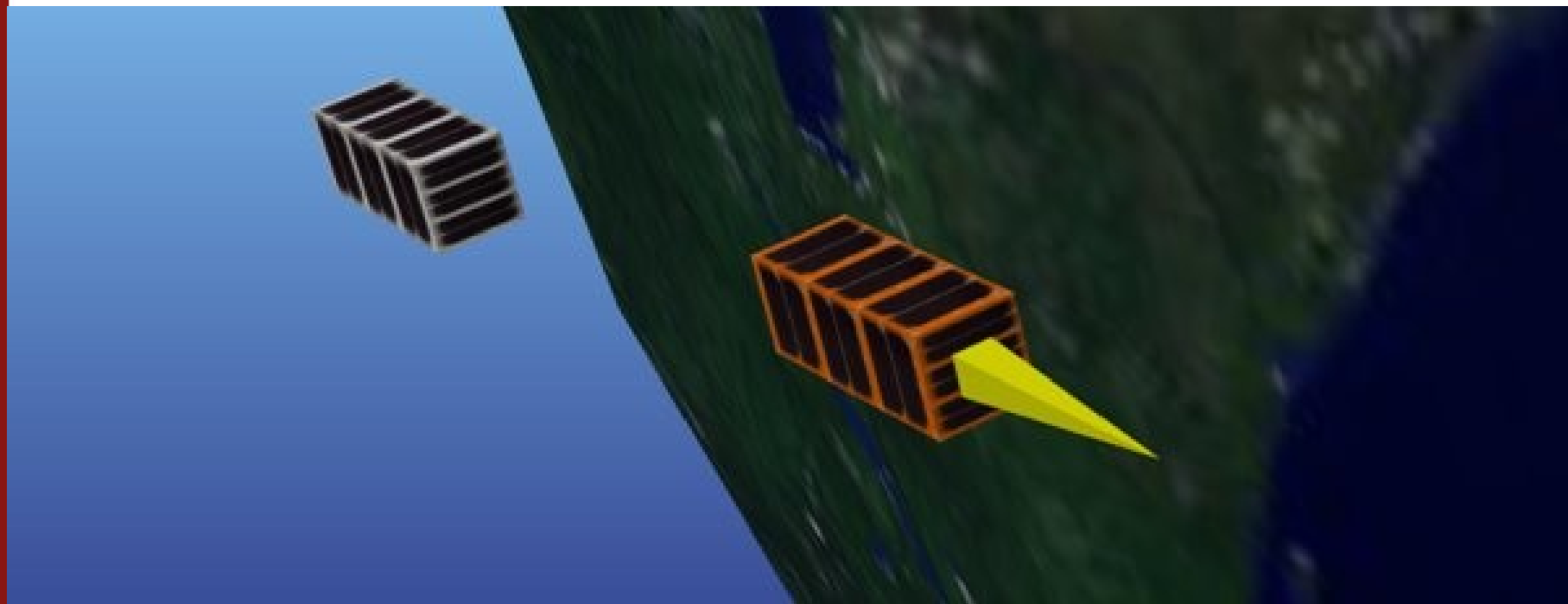
Simon Le Cleac'h and Zac Manchester

Robotic Exploration Lab

Motivation

Why L1-norm cost?

- Minimum-fuel, minimum-time
- Bang-off-bang control



Contribution

Solver:

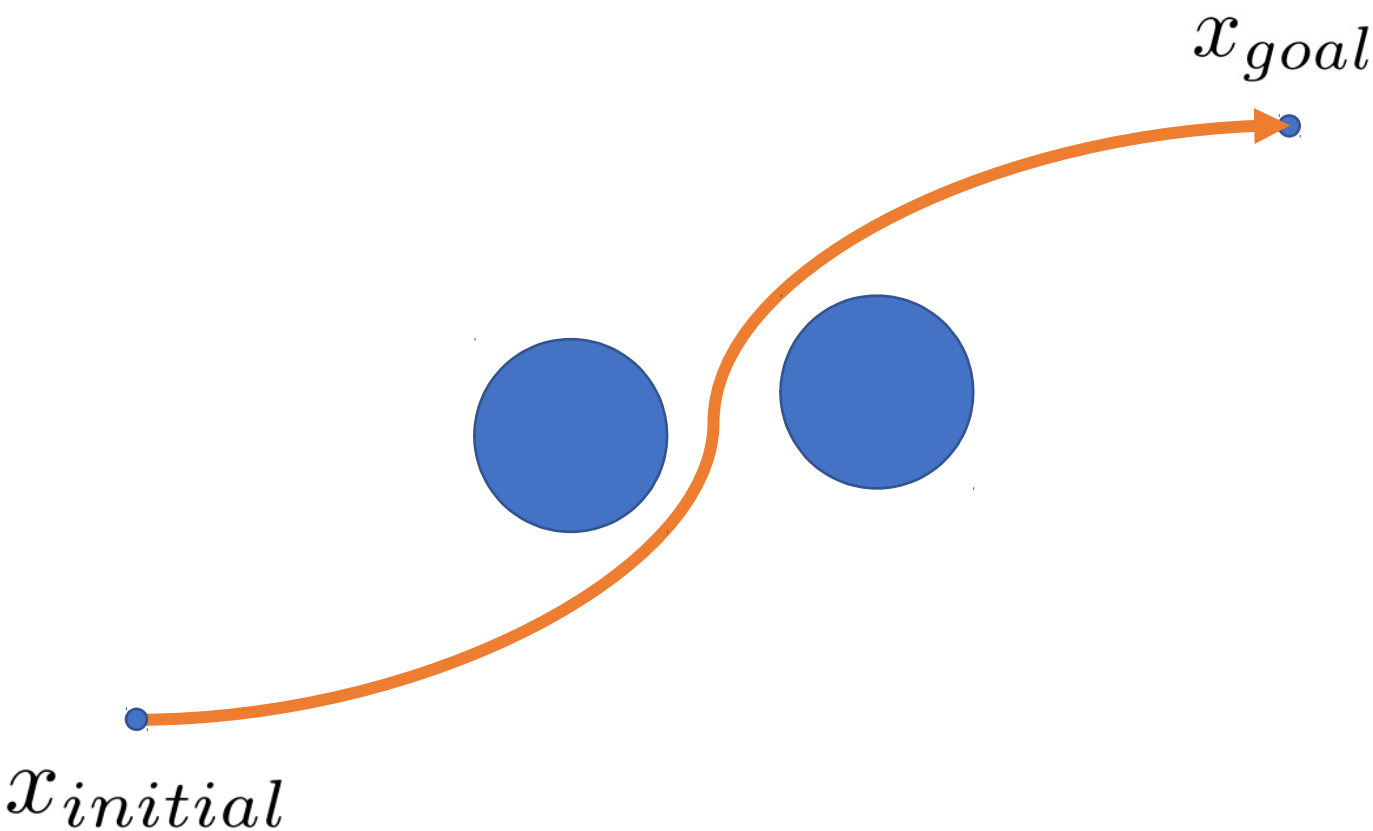
- Fast
- Low-memory footprint
- Nonlinear dynamics
- State and control constraints

Enables:

- In flight software implementation
- Embedded trajectory optimization

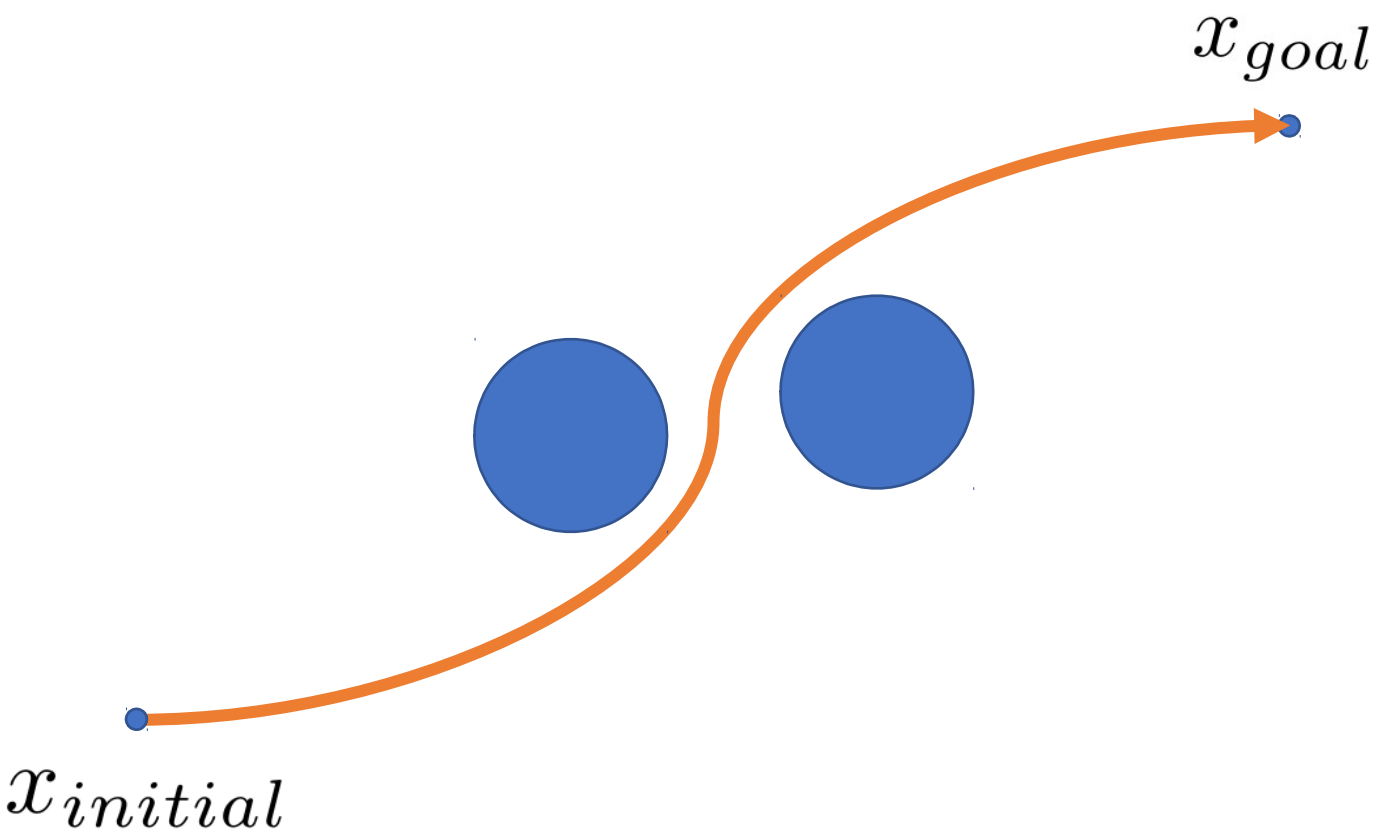
Trajectory Optimization

$$x_0(t), u_0(t) = argmin\ J(x, u)$$



Trajectory Optimization

$$x_{1:N}, u_{0:N-1} = argmin\ J(x_{0:N}, u_{0:N-1})$$



Trajectory Optimization

$$x_{1:N}, u_{0:N-1} = argmin\ J(x_{0:N}, u_{0:N-1})$$

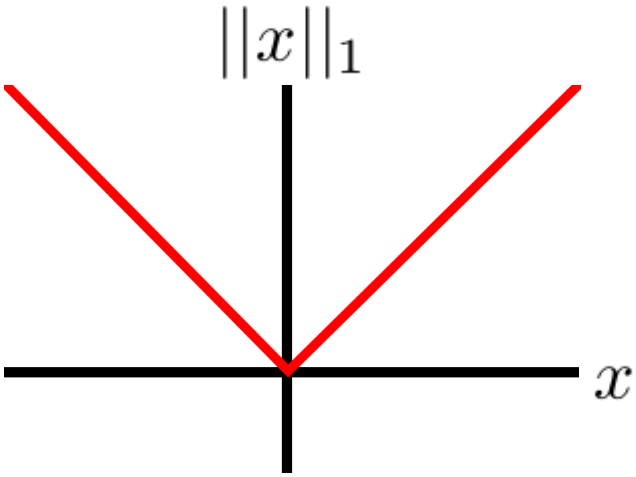
$$J(x_{0:N}, u_{0:N-1}) = l_f(x_N) + \sum_{k=0}^{N-1} l_k(x_k) + \boxed{\alpha||u_k||_1}$$

Trajectory Optimization

$$\underset{x_{1:N},\, u_{0:N-1}}{\text{minimize}} \quad J(x_{0:N}, u_{0:N-1}) = l_f(x_N) + \sum_{k=0}^{N-1} l_k(x_k) + \boxed{\alpha ||u_k||_1}$$

$$\text{subject to} \quad x_{k+1} = A_k x_k + B_k u_k$$

Nonsmooth cost function



ADMM

Problem form: f, g are convex

$$\begin{aligned} \underset{x,\,y}{\text{minimize}} \quad & f(x) + g(y), \\ \text{subject to} \quad & Ax + By = c \end{aligned}$$

Augmented Lagrangian:

$$\mathcal{L}(x, y, \lambda) = f(x) + g(y) + \lambda^T(Ax + By - c) + \frac{\rho}{2}||Ax + By - c||_2^2$$

ADMM

Augmented Lagrangian:

$$\mathcal{L}(x, y, \lambda) = f(x) + g(y) + \lambda^T(Ax + By - c) + \frac{\rho}{2}||Ax + By - c||_2^2$$

3 optimization steps:

$$x^{k+1} = argmin_x \ \mathcal{L}(x, y^k, \lambda^k)$$

$$y^{k+1} = argmin_y \ \mathcal{L}(x^{k+1}, y, \lambda^k)$$

$$\lambda^{k+1} = \lambda^k + \rho(Ax^{k+1} + By^{k+1} - c)$$

ADMM

Augmented Lagrangian:

$$\mathcal{L}(x, y, \lambda) = f(x) + g(y) + \lambda^T(Ax + By - c) + \frac{\rho}{2}||Ax + By - c||_2^2$$

3 optimization steps:

$$x^{k+1} = argmin_x \ \mathcal{L}(x, y^k, \lambda^k)$$ Minimization

$$y^{k+1} = argmin_y \ \mathcal{L}(x^{k+1}, y, \lambda^k)$$ Minimization

$$\lambda^{k+1} = \lambda^k + \rho(Ax^{k+1} + By^{k+1} - c)$$ Dual ascent

Trajectory Optimization

$$\underset{x_{1:N},\, u_{0:N-1}}{\text{minimize}} \quad J(x_{0:N}, u_{0:N-1}) = l_f(x_N) + \sum_{k=0}^{N-1} l_k(x_k) + \boxed{\alpha ||u_k||_1}$$

$$\text{subject to} \quad x_{k+1} = A_k x_k + B_k u_k$$

Method

$$\begin{aligned} \underset{x_{1:N},\, u_{0:N-1}}{\text{minimize}} \quad & J(x_{0:N}, u_{0:N-1}) \\ \text{subject to} \quad & x_{k+1} = A_k x_k + B_k u_k \end{aligned}$$

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Method

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Augmented Lagrangian:

$$\mathcal{L}(X, U, Y, \Lambda) = J(X, Y) + \sum_{k=0}^{N-1} \lambda_k^T (u_k - y_k) + \frac{\rho}{2} ||u_k - y_k||_2^2$$

Method

$$\begin{aligned} \underset{x_{1:N},\, y_{0:N-1}}{\text{minimize}} \quad & J(x_{0:N}, \underline{y_{0:N-1}}) \\ \text{subject to} \quad & x_{k+1} = A_k x_k + B_k u_k, \\ & \underline{u_k = y_k}. \end{aligned}$$

Augmented Lagrangian:

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Cost

Method

$$\begin{aligned} \underset{x_{1:N},\, y_{0:N-1}}{\text{minimize}} \quad & J(x_{0:N}, \underline{y_{0:N-1}}) \\ \text{subject to} \quad & x_{k+1} = A_k x_k + B_k u_k, \\ & \underline{u_k = y_k}. \end{aligned}$$

Augmented Lagrangian:

$$\mathcal{L}(X, U, Y, \Lambda) = J(X, Y) + \sum_{k=0}^{N-1} \lambda_k^T (u_k - y_k) + \boxed{\frac{\rho}{2} ||u_k - y_k||_2^2}$$

Penalty

Method

Augmented Lagrangian:

$$\mathcal{L}(X, U, Y, \Lambda) = J(X, Y) + \sum_{k=0}^{N-1} \lambda_k^T (u_k - y_k) + \frac{\rho}{2} ||u_k - y_k||_2^2$$

Alternating Direction Method of Multipliers (ADMM)

1. $$X^{i+1}, U^{i+1} = \text{argmin}_{X,U} \mathcal{L}(X, U, Y^i \Lambda^i)$$

2. $$Y^{i+1} = \text{argmin}_Y \mathcal{L}(X^{i+1}, U^{i+1}, Y, \Lambda^i)$$

3. $$\Lambda^{i+1} = \Lambda^i + \rho(U^{i+1} - Y^{i+1})$$

Method

Alternating Direction Method of Multipliers (ADMM)

1. $$X^{i+1}, U^{i+1} = \mathrm{argmin}_{X,U} \mathcal{L}(X, U, Y^i \Lambda^i)$$

Optimal control step <=> LQR, closed-form solution

$$\begin{aligned} \underset{X,U}{\text{minimize}} \quad & \mathcal{L}(X, U, Y^i, \Lambda^i), \\ \text{subject to} \quad & x_{k+1} = A_k x_k + B_k u_k \end{aligned}$$

Method

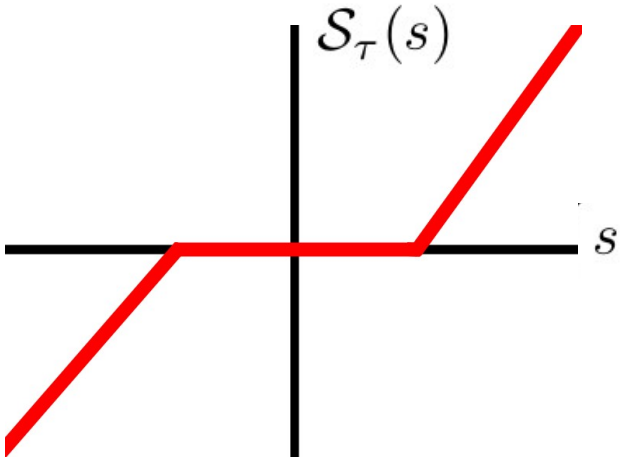
Alternating Direction Method of Multipliers (ADMM)

2. $$Y^{i+1} = \mathrm{argmin}_Y \mathcal{L}(X^{i+1}, U^{i+1}, Y, \Lambda^i)$$

Soft-threhold step <=> closed-form solution

$$Y^{i+1} = \mathcal{S}_{\alpha/\rho}\left(U^{i+1} + \frac{\Lambda^i}{\rho}\right)$$

$$\mathcal{S}_\tau(s) = \begin{cases} s - \tau & s > \tau \\ 0 & |s| \leq \tau \\ s + \tau & s < -\tau. \end{cases}$$



Method

Alternating Direction Method of Multipliers (ADMM)

3. $$\Lambda^{i+1} = \Lambda^{i} + \rho(U^{i+1} - Y^{i+1})$$

Dual ascent step <=> closed-form solution

Method

Alternating Direction Method of Multipliers (ADMM)

1. $$X^{i+1}, U^{i+1} = \mathrm{argmin}_{X,U} \mathcal{L}(X, U, Y^i \Lambda^i)$$

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2. $$Y^{i+1} = \mathrm{argmin}_{Y} \mathcal{L}(X^{i+1}, U^{i+1}, Y, \Lambda^i)$$

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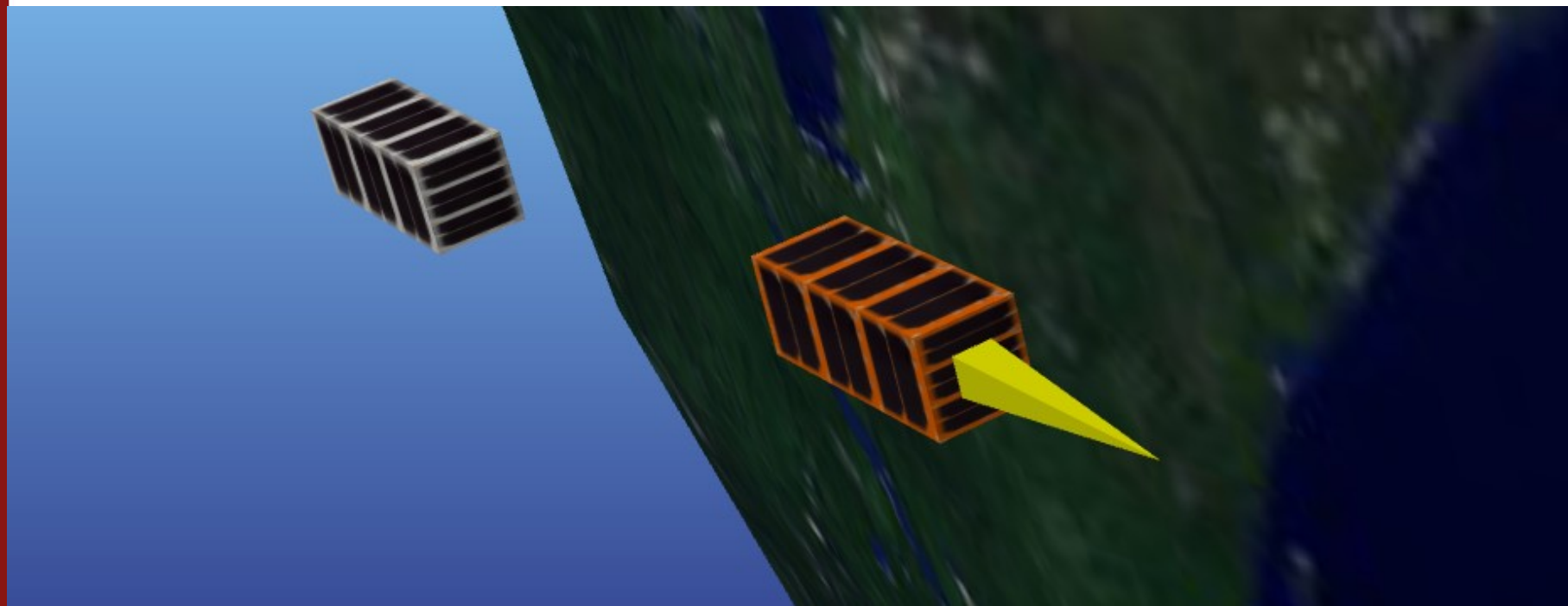
3. $$\Lambda^{i+1} = \Lambda^i + \rho(U^{i+1} - Y^{i+1})$$

   Dual ascent step <=> closed-form solution

Application

Spacecraft rendezvous problem

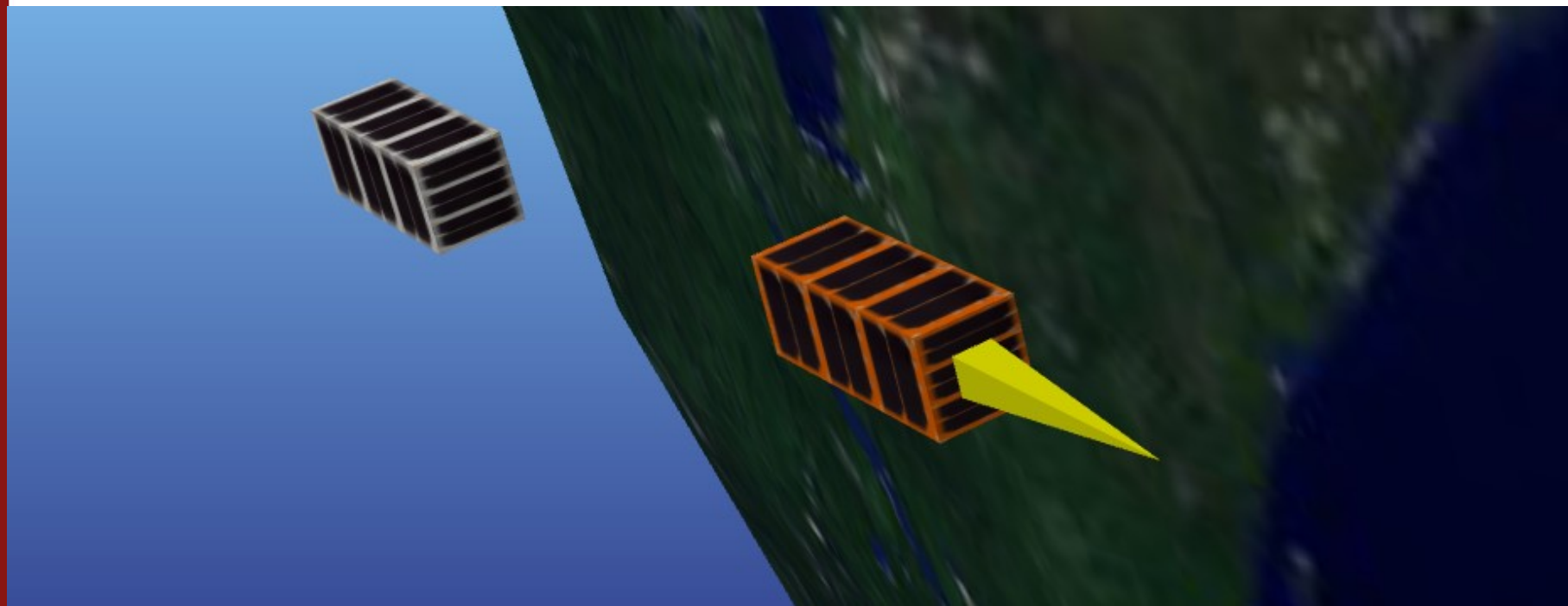
- Minimum-fuel
- Small satellite
- Pathfinder for Autonomous Navigation (PAN)



Application

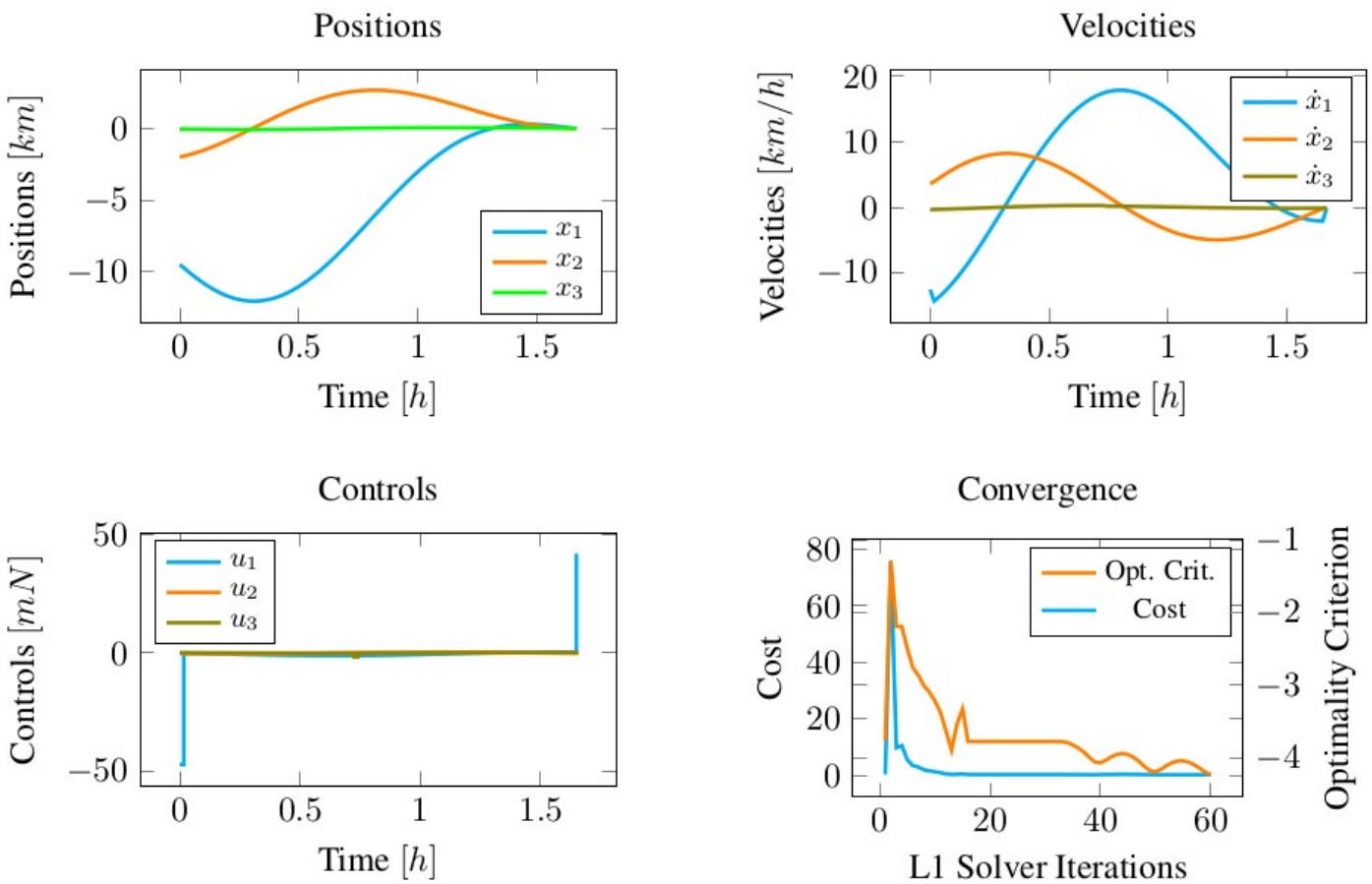
Spacecraft rendezvous problem

- Nonlinear dynamics (drag etc.)
- Unbounded control



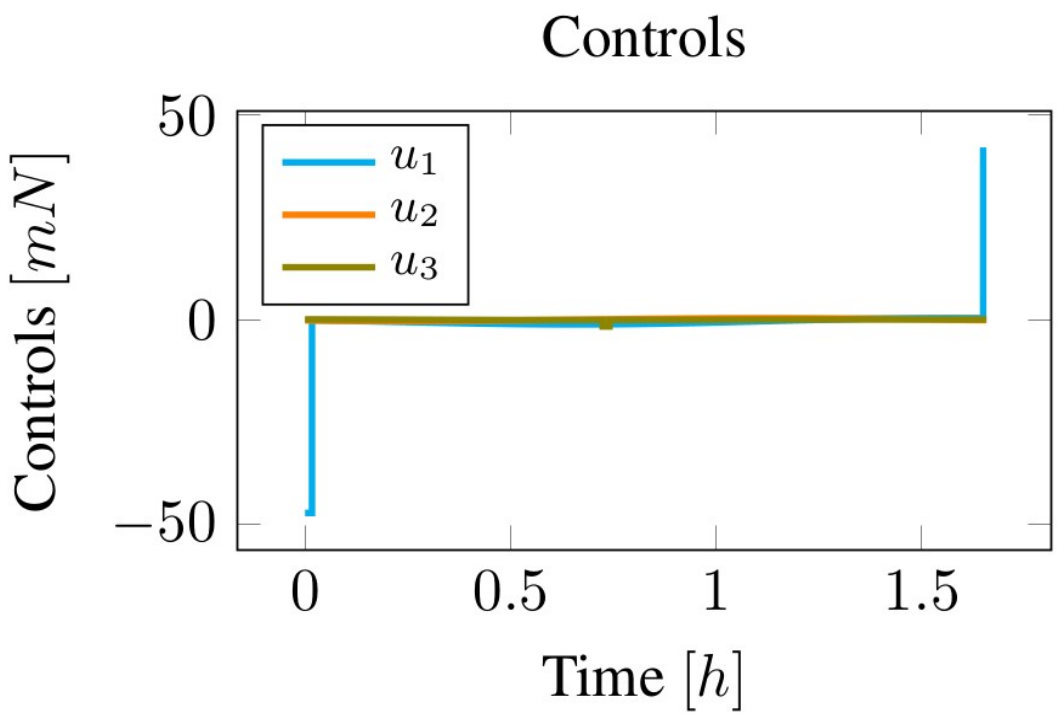
Results

- Nonlinear dynamics (drag etc.)
- Unbounded control



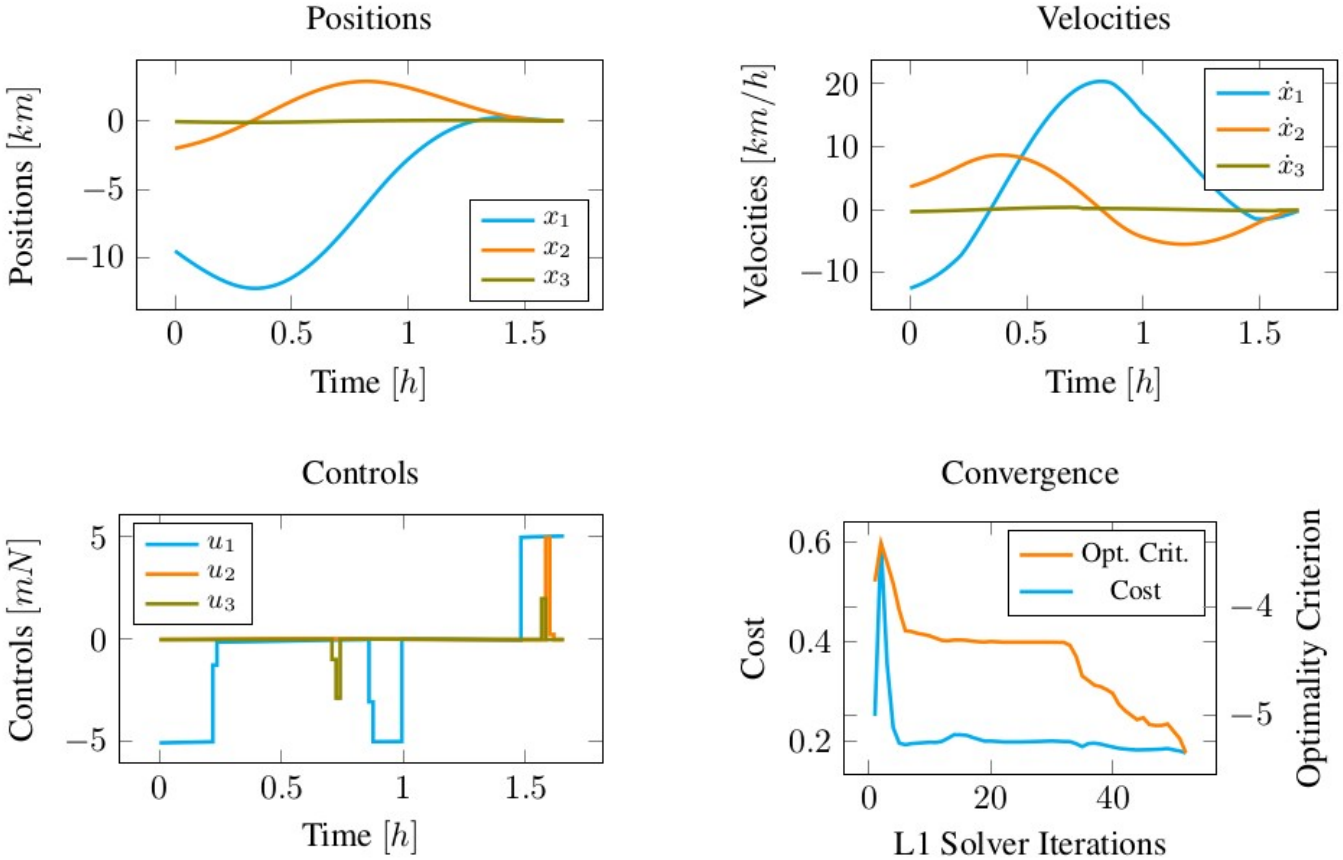
Results

- Nonlinear dynamics (drag etc.)
- Unbounded control
- Impulse control



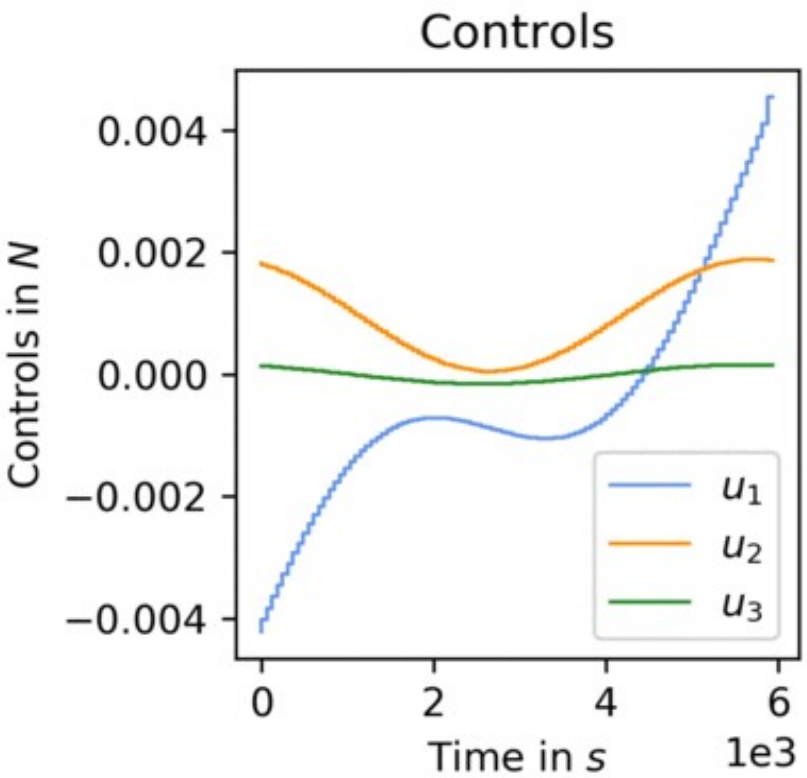
Results

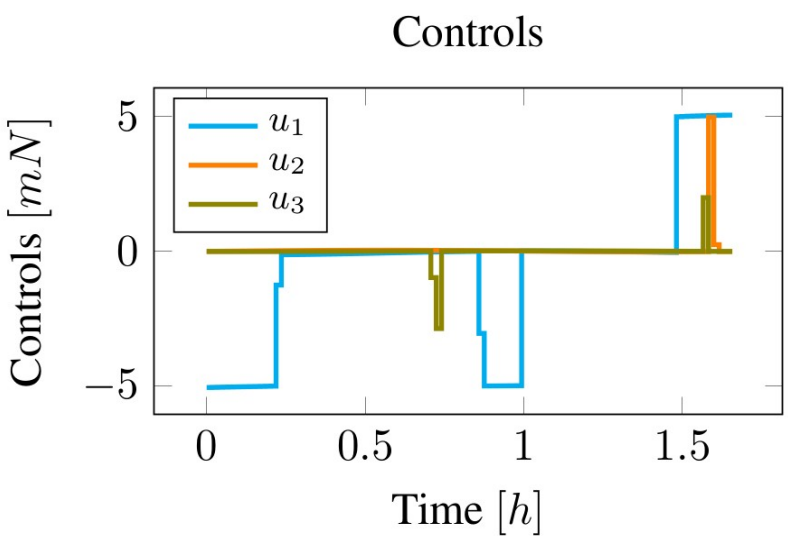
- Nonlinear dynamics (drag etc.)
- Bounded control



Results

- Nonlinear dynamics (drag etc.)
- Bounded control
- Bang-off-bang



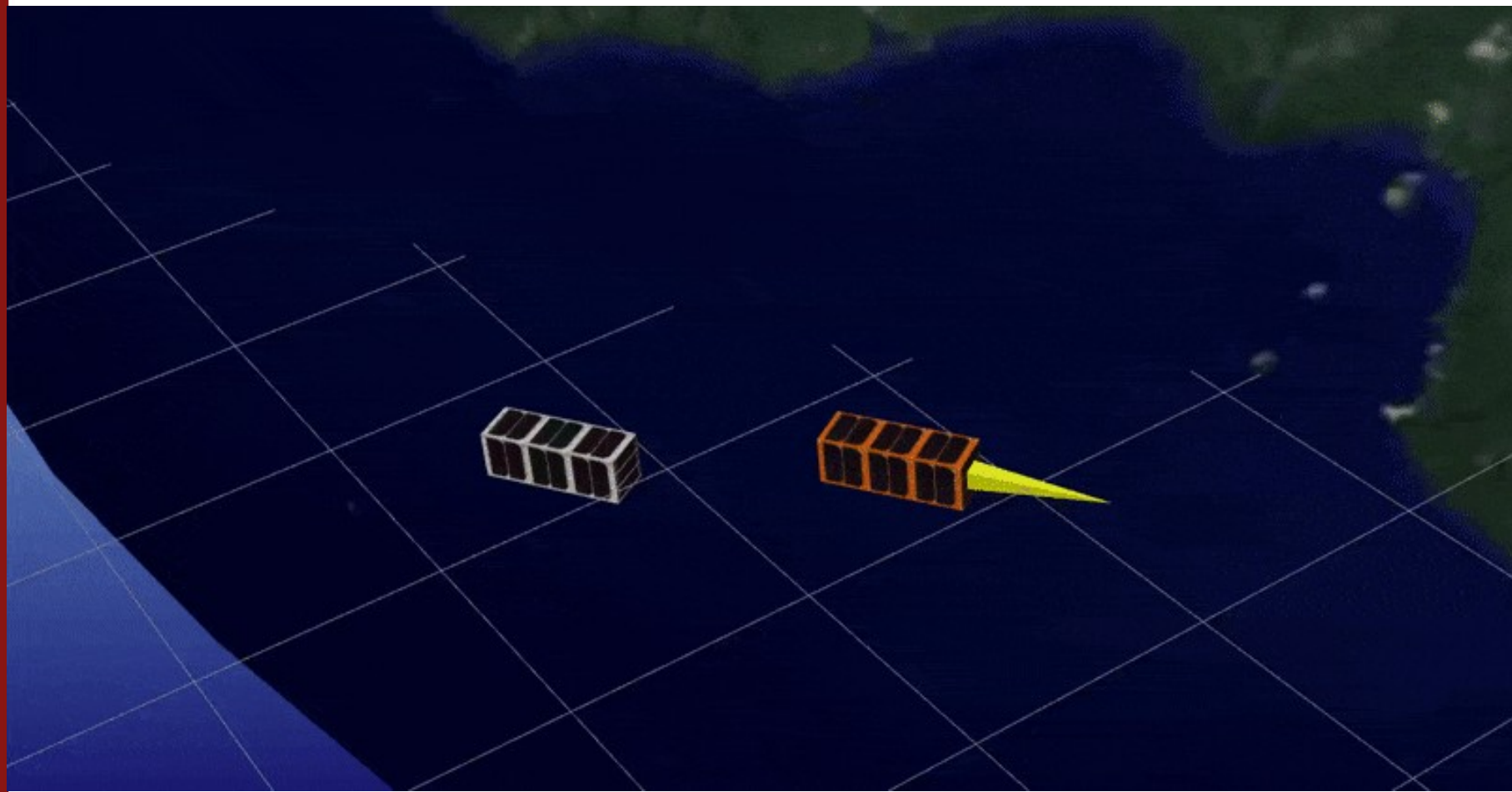


Results

Problem Type	Linear unconstrained	Linear constrained	Nonlinear unconstrained	Nonlinear constrained
Horizon (h)	1.7	1.7	1.7	1.7
Node Points (s)	100	100	100	100
ρ	1e-2	1e-1	1e-1	1e1
ϵ	4.5e-3	4e-4	6e-5	5e-6
Number LQR passes	86	404	120	206

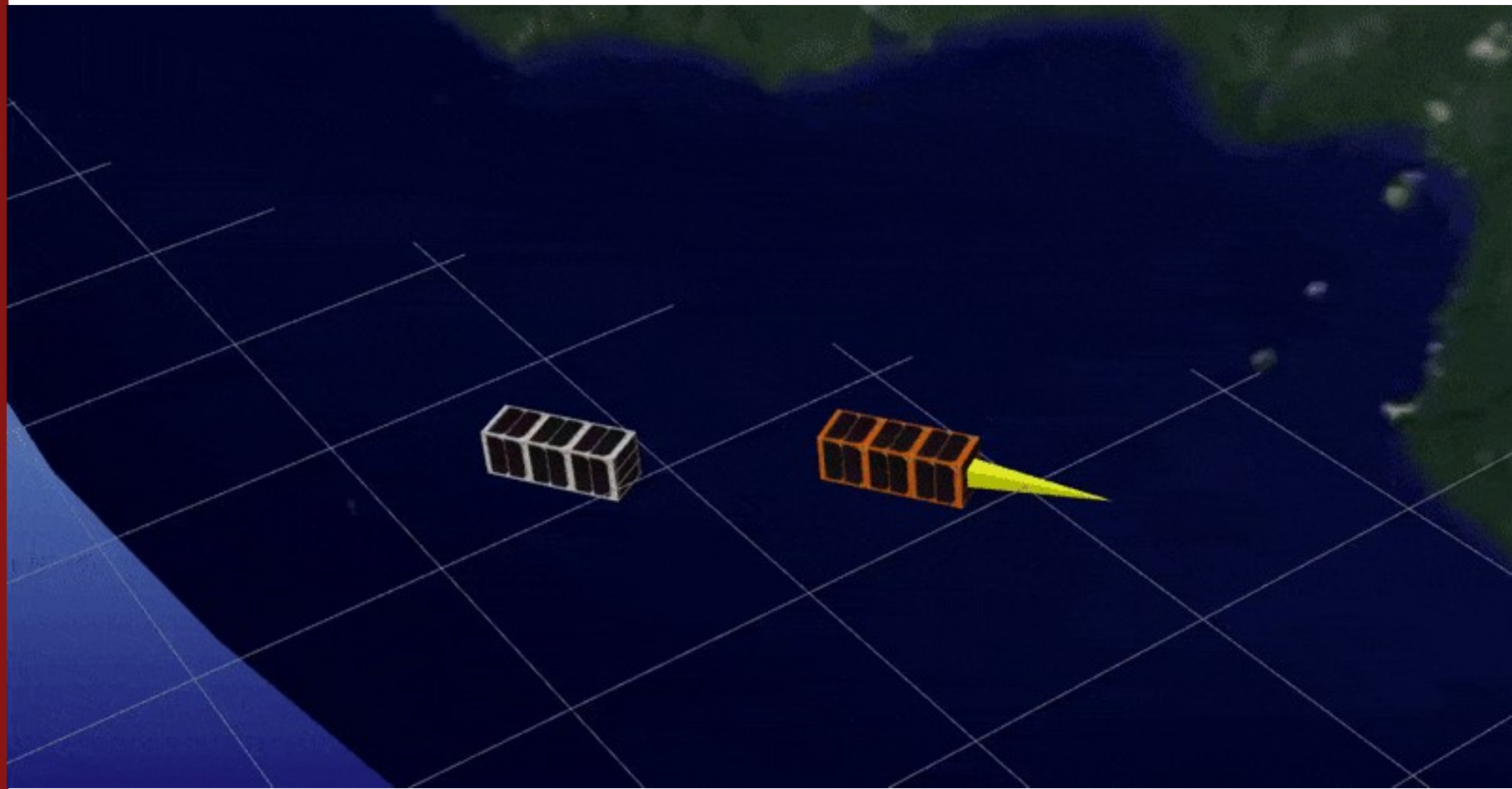
Spacecraft Rendezvous

L1 cost: Bang-off-bang



Spacecraft Rendezvous

Quadratic cost: smooth control



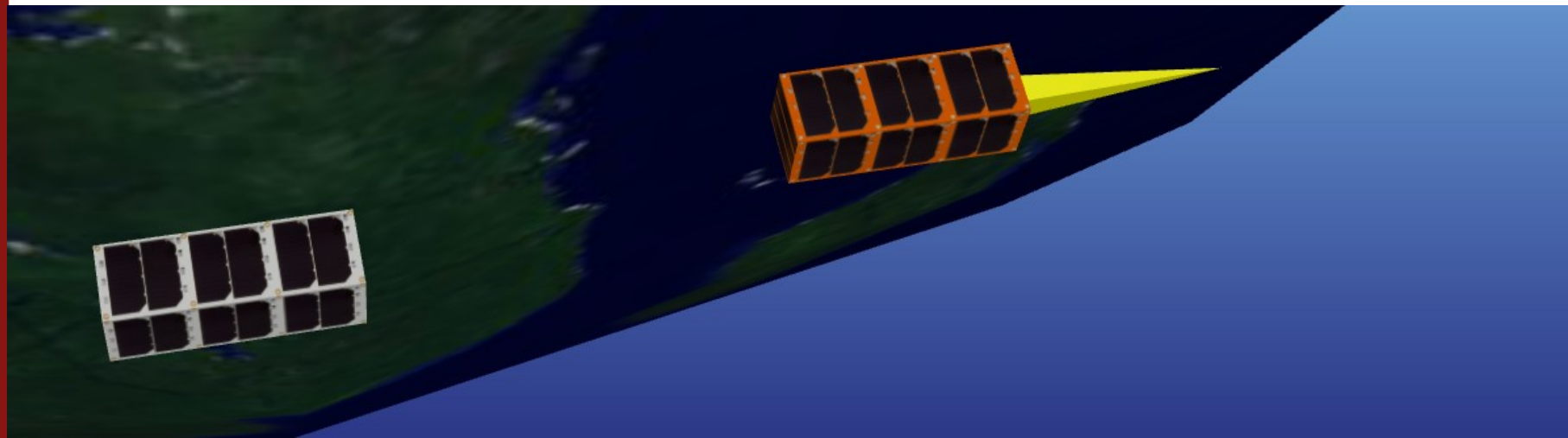
Conclusions

Solver: for L1 control cost problem.

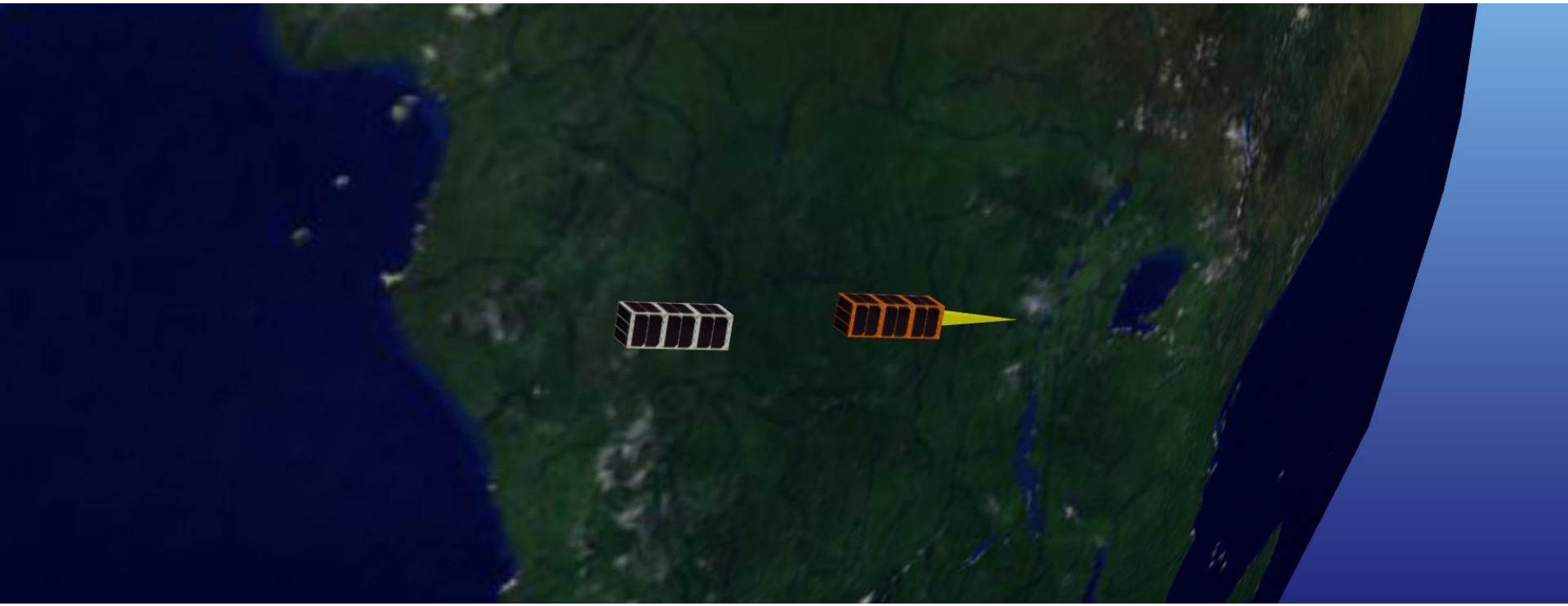
- Fast and low-memory footprint
- Broad range of applications in astrodynamics

Enables:

- In flight software implementation
- Small satellite rendezvous maneuver



Questions?



simonlc@stanford.edu

rexlab.stanford.edu

Algorithm

Algorithm 1 L1 Cost Optimizer

```
procedure L1CostOptimizer(x_0, U_initial; ρ, ε)
    X ← DynamicsRollout(x_0, U_initial)
    Y, Λ ← 0, 0
    repeat
        X, U ← OptimalControlUpdate(Y, Λ; ρ)
        Y ← SoftThresholdUpdate(X, U, Λ, ρ)
        Λ ← Λ + ρ (U − Y)
    until ||U − Y||_2 < ε
    return X, U
end procedure
```